Non-determinism in LLM-as-Judge Graders: An Empirical Reproducibility Note

Author: Hiroki Tamba (Tamba Research Academy) **Status:** Preprint / technical note — **v1.0 (report; pre-resolution)** **Origin:** Generalized from a finding first reported as `Japan-AISI/aisev` issue #25 (2026-06-03).

Abstract

LLM-as-judge ("grader") components are now standard in evaluation harnesses, including safety evaluations where a pass/fail verdict gates downstream decisions. A widespread implicit assumption is that setting the grader's sampling temperature to 0 makes grading *deterministic* and therefore *reproducible*. This note shows that assumption is wrong on two levels.

> **Uncontrolled defaults silently inject randomness.** In a real, publicly released evaluation harness (Japan AISI's `aisev`), the grader is invoked without setting `temperature` or `seed`. Because the OpenAI provider **omits** a `None` `temperature` from the request, the API applies its **default of `1.0`** (not a deterministic default), so the grader effectively runs **unseeded at temperature 1.0**. Items near the grader's decision boundary then flip pass/fail across otherwise identical runs.
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> **`temperature=0` is necessary but not sufficient.** Pinning the temperature to 0 removes the *rare* flips but not the hard ones: the actual grader (`gpt-4o`) still splits **2 of 7** items at `temperature=0`, and a Claude (Sonnet 4.6) grader independently reproduces a 2/7 split. Determinism of the *grader call* is not guaranteed by temperature alone, and the effect is not provider-specific.

We provide a small reproduction harness, report measured per-item disagreement rates, and recommend concrete mitigations: set `temperature=0` explicitly, set a `seed` where the provider supports it, run multiple `epochs` and **report grader variance instead of a single point estimate**, and surface a grader-disagreement rate as a harness health metric.

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1. Background

Modern evaluation pipelines increasingly replace human raters with an LLM "judge" that reads a model transcript and a rubric and emits a verdict (e.g. `correct` / `incorrect`, `pass` / `fail`). This is convenient and scalable, but it inherits a property practitioners often forget: **the judge is itself a stochastic model.** If the same transcript can be graded differently on two runs, then:

- benchmark scores are not reproducible,
- regression detection ("did this model get worse?") is confounded by grader noise, and
- safety gates that depend on a pass/fail threshold can flip for reasons unrelated to the model under test.

The common mitigation is "set temperature to 0." This note examines whether that mitigation is (a) actually applied in practice, and (b) sufficient when it is applied.

2. The discovered case: `aisev`

`aisev` is Japan AISI's open-source evaluation environment (Apache-2.0). All references below are to upstream commit `e0604d1e7997c2949e3cea40d219089dfd477fb4` (`Japan-AISI/aisev` `main` at 2026-04-16, the merge of PR #23: https://github.com/Japan-AISI/aisev/commit/e0604d1e7997c2949e3cea40d219089dfd477fb4). **This note does not redistribute aisev source code.** It describes the grader call path and points to the upstream files by path and commit hash, so each claim is verifiable against a fixed revision without vendoring any of the harness. Tracing the grader call path:

1. `scorer_provider.py` calls `model_graded_qa()` **without** passing `temperature` or `seed`.
2. This propagates into `inspect_ai`'s `GenerateConfig` with `temperature=None`.
3. The OpenAI provider **omits** a `None` temperature from the API request, so the API applies its **default of `1.0`** — full sampling, not deterministic decoding.
4. Net effect: the grader runs **unseeded, at temperature 1.0.**

Nothing in the harness signals to the user that grading is being performed under maximal sampling noise. The pass/fail numbers *look* like fixed measurements.

3. Empirical measurement

Method. We hold a fixed set of 7 deliberately borderline question/answer pairs (see `src/repro_disagreement_rate.py`) and grade each item repeatedly under two configurations: the harness's **default** (`temperature` and `seed` left unset, so the OpenAI provider omits the parameter and the API applies its default of 1.0 — N = 20 runs per item) and an explicit **`temperature=0`** (N = 10 runs per item). The grading prompt and the grade-extraction regex mirror aisev's `get_graded_qa_scorer` and `inspect_ai`'s default `model_graded_qa` instructions verbatim, so the verdict (`C`/`I`) is parsed exactly as aisev parses it. We compute the per-item *disagreement rate* — the fraction of runs whose verdict differs from the item's majority verdict.

Results — OpenAI `gpt-4o` grader (aisev's default). Three regimes appear: *stable* (no disagreement), *rare* (a single flip), and *strong* (near 50/50).

Item	default temp, N=20	`temperature=0`, N=10	Regime (default)	Disagreement (default)
item1	20 I / 0 C	10 I	stable	0.00
item2	20 I / 0 C	10 I	stable	0.00
item3	20 C / 0 I	10 C	stable	0.00
item4	11 I / 9 C	**6 I / 4 C**	**strong**	0.45
item5	19 I / 1 C	10 I	rare	0.05
item6	12 C / 8 I	**9 C / 1 I**	**strong**	0.40
item7	19 C / 1 I	10 C	rare	0.05

(Grader `gpt-4o`, aisev's default; the exact dated model snapshot was not captured in the run log. Raw outputs: `data/repro_raw_openai.json`.)

Under the default configuration, **4 of 7 items (4, 5, 6, 7) are non-reproducible** — two by a coin-flip margin (item4, item6) and two by a single flip (item5, item7). For these items the reported pass/fail is not a property of the answer under test; it is a draw from the grader's sampling distribution.

Pinning **`temperature=0` helps but does not close the gap.** It stabilises the two *rare* items (item5 and item7 become unanimous), yet the two *strong* items still flip — item4 splits 6/4 and item6 still yields a dissenting verdict (9/1) over just 10 runs. So on the actual aisev grader, `temperature=0` reduces non-reproducibility from **4/7 to 2/7**: necessary, but not sufficient.

4. The effect is not provider-specific

A natural objection is that §3 captures a quirk of one provider's grader. It does not. Running the identical 7-item protocol with a **Claude (Sonnet 4.6)** grader reproduces the same qualitative picture: 2 of 7 items are non-reproducible at the default temperature, and **still 2 of 7 at `temperature=0`** (item6: 8 I / 2 C; item7: 9 C / 1 I, over 10 runs).[1] The *specific* unstable items differ between graders — OpenAI's `temperature=0` instability falls on items 4 and 6, Claude's on items 6 and 7 — but the conclusion is identical: pinning the temperature does not make grading deterministic.

This falsifies the operational belief that `temperature=0 ⇒ deterministic grading`. Residual non-determinism in production LLM serving can arise from sources unrelated to the sampling temperature, including:

- batching / request-coalescing effects on the serving side,
- mixture-of-experts routing variability,
- non-associative floating-point reductions across hardware/kernels,
- provider-side load balancing across non-identical model replicas.

The practical consequence: **`temperature=0` reduces, but does not eliminate, grader disagreement.** Reproducibility must be *measured*, not assumed.

5. Why this matters

For ordinary capability benchmarks, grader noise inflates variance and can change rankings of closely-spaced models. For **safety** evaluations the stakes are higher: a pass/fail verdict near the boundary may gate a deployment decision, a red-team sign-off, or a public claim about a guardrail's coverage. If that verdict is a coin flip, the evaluation is reporting noise as a safety property.

6. Recommended mitigations

1. **Set `temperature=0` explicitly** at the grader call site. Never rely on the provider's `None` default — it may mean `1.0`.
2. **Set `seed`** where the provider supports it, and record it in the run metadata.

3. **Run `epochs > 1`** for the grader and **report variance / a confidence interval**, not a single point estimate.
4. **Treat near-boundary items as a first-class signal.** Surface a per-run *grader-disagreement rate* as a harness health metric; a high rate means the score is not trustworthy regardless of its central value.
5. **Log the effective grader config** (model, temperature, seed) into the results artifact so that "what temperature was the judge at?" is answerable after the fact.

7. Reproduction

The reproduction harness is `src/repro_disagreement_rate.py`. It mirrors aisev's `get_graded_qa_scorer` prompt and `inspect_ai`'s default grade-extraction verbatim, grades the 7-item set under both the default and `temperature=0` conditions, and reports the per-item disagreement rate. The raw grader outputs backing §3–§4 are in `data/`. See `README.md` for setup and the exact run counts.

8. Related work and positioning

LLM-as-judge reliability. Using an LLM to score model outputs is now standard practice (Zheng et al., 2023). Most of this literature characterizes *systematic* biases of the judge — position bias, verbosity bias, and self-preference (Wang et al., 2023) — that is, skews that persist *across* runs. The finding here is on an orthogonal axis: *run-to-run* non-determinism, where the same judge returns different verdicts for the identical (transcript, rubric) pair on repeated calls. The two concerns are complementary: a debiased judge can still be irreproducible, and a perfectly reproducible judge can still be biased.

Eval-harness grader configuration. Frameworks such as Inspect AI (UK AISI) expose the grader as a configurable model call (`model_graded_qa`, `GenerateConfig`), including `temperature` and, where the provider supports it, `seed`. The gap this note highlights is not in a framework's capabilities but in its *defaults*: when these knobs are left unset, the underlying provider may silently sample (OpenAI maps `temperature=None → 1.0`), so a harness can report grades produced under full sampling noise without ever signalling it. This is a configuration-level reproducibility hazard, not a modeling one.

Residual non-determinism at `temperature=0`. Even under greedy decoding, production LLM serving is generally not bit-reproducible: request batching, mixture-of-experts routing, and non-associative floating-point reductions across kernels and hardware can change the argmax for borderline tokens; recent analysis attributes temperature-0 nondeterminism primarily to a lack of *batch invariance* in core kernels, where numerics shift with batch size and sequence slicing (He et al., 2025). The cross-model check in §4 is consistent with this — a Claude grader at `temperature=0` still flipped 2/7 items — and motivates the operative recommendation of this note: *measure grader disagreement; do not assume it away.*

Pointers. - Original report: `Japan-AISI/aisev` issue #25 — *"Reproducibility Improvement Proposal — Grader's Pass/Fail Fluctuates Due to Uncontrolled Temperature/Seed."* https://github.com/Japan-AISI/aisev/issues/25 - Inspect AI (UK AISI): `GenerateConfig`, `model_graded_qa`.

References.

1. L. Zheng, W.-L. Chiang, Y. Sheng, et al. "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena." *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023. arXiv:2306.05685, DOI: 10.48550/arXiv.2306.05685.
2. P. Wang, L. Li, L. Chen, et al. "Large Language Models are not Fair Evaluators." 2023. arXiv:2305.17926, DOI: 10.48550/arXiv.2305.17926.
3. H. He et al. "Defeating Nondeterminism in LLM Inference." Thinking Machines Lab, 2025. https://thinkingmachines.ai/blog/defeating-nondeterminism-in-llm-inference/

Scope of this deposit

To stay within upstream licensing and good-citizen norms, this deposit contains **only the author's own artifacts**:

- the reproduction harness (`src/`),
- the measured results and their analysis (this note), and
- a written description of the upstream grader call path that references `Japan-AISI/aisev` **by file path and commit hash** rather than copying its source.

It does **not** redistribute any portion of the `aisev` codebase. Readers reproduce the upstream context by checking out the referenced commit of the upstream repository directly.

Versioning

This artifact follows a two-version plan, archived under a single Zenodo *concept* DOI with a distinct *version* DOI per release:

- **v1.0 (this release) — report.** Documents the finding and the measured non-determinism as of upstream commit `e0604d1` (`main` at 2026-04-16, PR #23), before any fix is adopted.
- **v2.0 (planned) — resolution.** If the proposed mitigations are adopted upstream, v2 will add a short "Resolution" section recording the change (the upstream PR / commit and the post-fix disagreement rate). Citations to the concept DOI will resolve to the latest version; citations to a version DOI remain pinned.

See `CHANGELOG.md`.

Acknowledgments and tooling

The empirical finding, reproduction code, and analysis are the author's own. Drafting of this note was assisted by an LLM (Claude); all claims and data were verified by the author.

How to cite

See `CITATION.cff`. Once archived, cite by the Zenodo DOI minted on release.

Cross-model check: grader `claude-sonnet-4-6`, same 7-item set, N = 20 at the default temperature and N = 10 at `temperature=0`. Non-reproducible items were {6, 7} under both conditions (default — item6 14 I / 6 C, item7 15 C / 5 I; `temperature=0` — item6 8 I / 2 C, item7 9 C / 1 I). The `temperature=0` batch is small and is meant only to show that disagreement does not vanish, not to estimate a precise rate. Raw outputs: `data/repro_raw_anthropic.json`. ■